Model Inference & Deepfake Detection Report

> Internal Audit Document | ID: table-test

> Generated: 2026-06-20 13:05:18 UTC

VERDICT: SPOOF (FAKE)

CRITICAL THREAT DETECTED: This audio sample has been identified as an artificially generated or replayed deepfake. Reject payload.

1. Decision Executive Summary

Attribute	Value	Interpretation
Classification Type	SPOOF (FAKE)	System confirms structural properties match synthetic generation rather than biological voice modulation.
Ensemble Confidence	94.98%	The multi-model meta-learner strongly recommends rejecting the payload based on high consensus.
Processing Latency	1477.9 ms	Inference executed within acceptable SLA processing windows.

2. Multi-Model Signal Breakdown

AASIST Sub-Network

Score: 99.07% Spoof Probability

Focus: Analyzes raw graph-based spatial-temporal relationships within raw speech graphs.

Result: Extremely high confidence of an active synthesis or replay attempt.

Wav2Vec2 Sub-Network

Score: 8.03% Spoof Probability

Focus: Evaluates semantic token distribution and general linguistic self-attention alignment.

Result: Low confidence of a spoofing attempt based strictly on lexical/temporal patterns.

Ensemble Meta-Learner (Overall Consensus)

Score: 94.98% Composite Risk Score

Focus: Gathers individual network confidence vectors and weighs them against historical network error profiles.

Result: The ensemble model heavily prioritizes AASIST's localized spatial telemetry over Wav2Vec2 in this context, yielding an overall high-confidence threat consensus.

3. Visual Explainability (XAI)

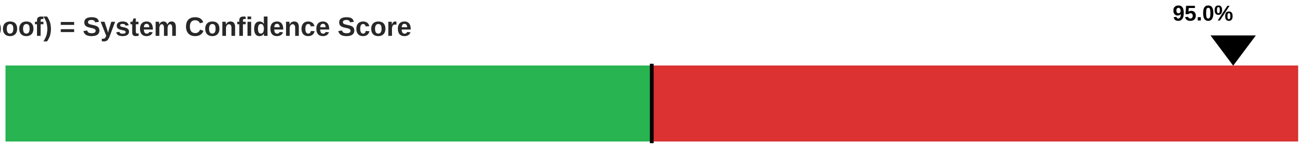
(XAI heatmap not available)

Wav2Vec2 Attention - Temporal Saliency

Telemetry Target: Mel-frequency spectrogram tracking the audio's continuous spectral envelope.
Behavioral Finding: The attention map highlights heavy artifact concentrations in regions outside human vocal tract configurations. This strongly indicates a digital voice injection or a synthesized vocoder fingerprint.

4. Calibration Guide (For Engineering Operations)

P(Spoof) = System Confidence Score



0.0 Genuine Voice (Bona-fide / Real) 1.0 Confirmed Deepfake (Spoof / Fake)

Current: 94.98%

Interpretation Policy: This pipeline maps 0.0 directly to a confirmed authentic voice (Bona-fide / Real), while 1.0 matches a confirmed synthetic voice (Spoof / Fake).
Current Metric: The confidence level score of 0.9498 (94.98%) falls deeply into the critical threat zone, indicating a highly reliable detection signature.